

BACKGROUND PLACEHOLDER
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Quality Control (QC) Audit Report

Software Requirements Specification (SRS)

Smart Contract Enabled Pharmaceutical Supply Chain System

Veridion Systems Validation Labs

March 18, 2026

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Table 1: QC Team – Veridion Systems Validation Labs

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1 Executive Summary

This report presents the results of a Quality Control (QC) validation of the Software Requirements Specification (SRS) for the **Smart Contract Enabled Pharmaceutical Supply Chain System**. The objective of this QC audit is to identify defects that remain in the SRS after the completion of Eliciting, Analysing, and Specifying.

The audit focused on three principal categories of SRS defects expected at the Validating stage:

- **Authorship defects** – where intended meanings are not expressed clearly enough.
- **Omission defects** – where required scenarios, conditions, or supporting content are missing.
- **Contradiction defects** – where different parts of the SRS imply inconsistent behaviour.

The inspection found that the SRS is generally well-structured and covers most core functional areas. However, a number of issues were detected including contradictory requirements (MFA scope, blockchain confirmation time), missing exception-handling scenarios (blockchain failure, invalid QR codes, password recovery), vague statements, and typographical errors. A glossary is absent, and some cross-references are inconsistent.

These findings indicate that the SRS is **conditionally acceptable**, subject to the customer's review of the identified defects. No recommendations for correction are provided; the customer will determine any necessary actions.

2 Introduction

2.1 Purpose

This Quality Control (QC) Audit Report presents the findings of an independent inspection of the Software Requirements Specification (SRS) prepared for the **Smart Contract Enabled Pharmaceutical Supply Chain System**. The purpose of this QC activity is to identify defects remaining in the SRS work product before the project progresses into downstream stages such as system design and implementation.

In accordance with the validating stage of Requirements Engineering, QC focuses on defect detection in the **requirements work product itself**. The QC team records findings but does not propose fixes; the customer determines any corrective actions.

2.2 Document Conventions

This document is an audit report rather than a replacement SRS. It records detected defects and explains their implications.

Audit findings use the format QC-FND-XXX.

Error detection instances use the format QC-ERR-XXX.

2.3 Intended Audience and Reading Suggestions

This report is intended for the Medicare Plus requirements engineering team, project management personnel, and relevant stakeholders involved in SRS review and approval.

Readers should begin with the Executive Summary and Sections 1–3, then proceed to the QC Checklist, Error Detection, and detailed Findings.

2.4 Audit Scope

The audit covers the SRS as a QC exercise. The scope includes:

- detection of authorship defects (ambiguity, vagueness, grammatical errors)
- detection of latent contradictions
- detection of omission of topics (missing scenarios, exception paths)
- checking codification and traceability coherence
- checking structural, editorial, and presentation consistency

3 Audit Basis

3.1 Basis of Evaluation

This QC audit was conducted on the understanding that the SRS should already be in reasonably strong condition after completion of Eliciting, Analysing, and Specifying. The purpose of QC at this stage is therefore not to rewrite the SRS, but to detect remaining defects still likely to be present in the work product.

3.2 Nature of QC

Quality Control focuses on inspecting the SRS work product itself. The QC team identifies defects but does not directly amend the SRS. Instead, the audit team produces a report that identifies defects for consideration by the customer.

3.3 Kinds of Errors Pursued

The QC audit focuses on detecting the following major defect categories expected in the SRS at the validating stage:

- **Indiscriminate side effects of authorship** – where intended meaning is clear to the writers but not expressed with sufficient precision.
- **Latent contradictions** – where requirements, models, or sections conflict with one another.
- **Omission of topics** – where expected requirements content, edge conditions, or supporting information is absent.
- **Traceability coherence issues** – where codification, source mapping, or cross-references are incomplete or unclear.
- **Editorial and structural inconsistencies** – where formatting, section naming, or presentation patterns weaken readability or professionalism.

3.4 Severity Classification

Findings are categorised according to severity:

- **High** – defect that may significantly affect interpretation, implementation, or expected system behaviour.
- **Medium** – defect that may introduce ambiguity or reduce clarity without fundamentally changing requirement intent.
- **Low** – minor editorial or structural issue that does not materially affect requirement meaning.

3.5 Documents Reviewed

The audit considered the following project materials as supporting references:

- Software Requirements Specification (SRS) for the Smart Contract Enabled Pharmaceutical Supply Chain System, version as of 6 March 2026.
- Project Specification (assumed background).
- Relevant requirements models (context diagram, sequence diagram, architecture diagram).
- Codification and iRTM sections.
- Quality model and change mitigation sections.
- Appendices and supporting traceability artefacts.

4 QC Plan

4.1 Target Workproduct

The target workproduct for this QC audit is the released **Software Requirements Specification (SRS)** for the Smart Contract Enabled Pharmaceutical Supply Chain System project.

4.2 Context and Goals

The SRS is the main requirements work product produced after Analysing and Specifying. Since downstream design and implementation rely on the SRS as the controlling requirements baseline, the QC audit is conducted to provide confidence that the specification is sufficiently clear, complete, and internally consistent.

The goals of this QC audit are:

- to determine whether the SRS clearly and consistently represents the intended requirements.
- to identify material defects before downstream design and implementation begin.
- to document defects for customer consideration.

4.3 Access and Audit Conditions

The QC team reviewed the released SRS and supporting project artefacts made available for audit. Access was limited to materials needed to inspect the work product and its supporting traceability and modelling references.

4.4 Inspection Techniques Used

The following investigative techniques, drawn from industry best practices and the lecture material, were applied during the QC audit:

1. **Close document inspection** – detailed reading of requirement clauses and supporting sections.
2. **Perspective-based reading** – reviewing the SRS from likely reader viewpoints such as developer, tester, project manager, and compliance officer.
3. **Cross-section contradiction checking** – comparing different SRS sections and models to detect hidden conflicts (e.g., performance targets, security scopes).
4. **Codification and traceability verification** – checking requirement IDs, source links, iRTM mappings, and cross-references.
5. **Model-to-text consistency inspection** – ensuring models (diagrams) and textual requirements express compatible meanings.
6. **Omission mapping against expected SRS topics** – checking whether important topics, edge cases, and exception paths are sufficiently covered (e.g., failure handling, recovery processes).
7. **Scenario-based reading** – testing the SRS against realistic supply-chain scenarios (order placement, product verification, blockchain outage).
8. **Assumption reading** – identifying assumptions that contradict documented requirements or have been rendered redundant.
9. **Double-checking** – verifying dates, facts, figures, and proper names for consistency.
10. **Spellcheck and grammar review** – using automated tools plus manual reading to catch typographical errors.

4.5 Work Breakdown

The QC audit team consisted of four members, with one member also serving as the QA auditor (not reflected in this QC report). The work was allocated as follows:

Role	Assigned Member	Primary Responsibility
QC Audit Lead	Muhammad Ridwan Putra Jasni	Coordinate audit activities, consolidate findings, ensure consistency of final report, apply cross-section contradiction checking and assumption reading.
Requirements Reviewer	Lin Yuhao	Inspect operational, technology, and quality-related sections for ambiguity, contradiction, and omission; perform perspective-based reading (developer, tester) and scenario-based reading.
Traceability Reviewer	Muhammad Mikhail Bin Mazlan	Review codification, iRTM references, source mappings, and section-to-section coherence; double-check all identifiers and cross-references.
Editorial and Structure Reviewer	Kannan s/o Rajamohan	Review document structure, section naming, formatting consistency, and overall presentation quality; conduct spellcheck and grammar review, verify figures and tables.

Table 2: QC audit team work allocation

4.6 Indicative Timeline

The audit was conducted over a single day, with the following estimated timeline:

Stage	Activity	Indicative Duration
Stage 1	Individual inspection of assigned SRS sections using selected techniques (close reading, perspective-based, scenario-based)	3 hours
Stage 2	Consolidation meeting and de-duplication of identified findings	1 hour
Stage 3	Severity review and documentation of findings	1 hour
Stage 4	Final report formatting, checking, and team sign-off	1 hour

Table 3: Indicative QC audit timeline

4.7 Reporting

The QC team records findings in this report. The audit team does not directly amend the audited SRS.

5 QC Checklist

The following checklist was used by the QC audit team to inspect the SRS work product. The checklist items represent critical aspects of a high-quality SRS, derived from industry standards and lecture guidelines.

Table 4: QC inspection checklist

Item	Focus	QC Checklist Item	Inspection Result
1. Introduction Section			
QC1	Completeness	Does the Introduction explicitly define the system purpose, target user base, and operational boundaries (in-scope vs. out-of-scope)?	Satisfied
QC2	Completeness	Does the Introduction state the specific business problem and identify measurable business goals or KPIs?	Partially satisfied
QC3	Consistency	Is the system name, project title, and version number identical across title page, headers/footers, and body?	Satisfied
2. Glossary / Definitions Section			
QC4	Completeness	Are all domain-specific terms, acronyms, and technical abbreviations defined in a Glossary?	Not satisfied (missing)
QC5	Relevance	Does every term defined in the Glossary appear at least once in the body?	Not applicable
QC6	Consistency	Is each system role or entity referred to consistently using the same term throughout?	Satisfied
3. Overall Description Section			
QC7	Completeness	Are all user classes identified with their technical expertise, access privileges, and responsibilities?	Satisfied
QC8	Completeness	Does the SRS specify the intended operational environment (OS, browsers, hardware, network)?	Satisfied
QC9	Consistency	Are all listed dependencies and constraints adhered to and not contradicted by later requirements?	Partially satisfied
4. Functional Requirements Section			
QC10	Correctness	Does each requirement describe a single behaviour using precise, testable language (avoiding vague terms)?	Partially satisfied
QC11	Testability	Does each functional requirement define trigger inputs, expected behaviour, and output/error state?	Partially satisfied
QC12	Consistency	Is every requirement assigned a unique identifier following a consistent naming convention?	Satisfied
5. Interface Requirements Section			
QC13	Completeness	Does the Interface section describe purpose, data exchanged, and communication method for each external interface?	Partially satisfied
QC14	Correctness	Is the trigger condition, expected data payload, and error handling defined for each interface?	Partially satisfied
QC15	Consistency	Does every external interface listed correspond to at least one functional requirement?	Satisfied
6. Non-Functional Requirements Section			
QC16	Testability	Are all NFRs quantified with measurable limits (e.g., response time ≤ 2 sec for 95% of queries)?	Partially satisfied
QC17	Completeness	Does the SRS address all relevant NFR categories (performance, security, reliability, usability, maintainability)?	Satisfied
QC18	Consistency	Do the NFRs logically align with system constraints (e.g., storage capacity consistent with data volume)?	Satisfied

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Item	Focus	QC Checklist Item	Inspection Result
7. Addenda / Supporting Artifacts			
QC19	Traceability	If an iRTM is included, does it map requirements to business goals and validation artifacts?	Satisfied
QC20	Correctness	Does every diagram include a descriptive title, legend, and standard notation accurately representing the system?	Partially satisfied
QC21	Consistency	Do all cross-references, figure numbers, and table citations point to correct existing sections/artifacts?	Partially satisfied

6 Error Detection

The QC audit identified instances of three principal QC error types: authorship, omission, and contradiction. Each detected instance is recorded with its location and characteristic.

Error ID	Error Type	Location	Characteristic of Error
QC-ERR-001	Contradiction	Section 1.4.1 (In-Scope Deliverables, Authentication & Security) vs Section 3.2 FR-2.6	Section 1.4.1 requires MFA for "all user accounts", while FR-2.6 requires MFA only for "all administrative accounts". These statements directly conflict on the scope of MFA enforcement.
QC-ERR-002	Authorship	Section 5.1 NFR-4 and NFR-7	Duplicate word "shall" appears in both requirements: "The system shall shall complete 95% of API responses within 2 seconds" and similar for database queries. This is a typographical error.
QC-ERR-003	Omission	Section 3.2 (User & Role Management)	FR-2.5 specifies account lockout after failed attempts, but no requirement exists for password reset or account recovery flow. This is a missing functional capability.
QC-ERR-004	Authorship	Section 3.3 FR-3.9	The term "periodic" is used without definition: "submit periodic GPS updates". The frequency or condition for updates is not specified, making the requirement ambiguous.
QC-ERR-005	Omission	Section 3.1 (Blockchain Infrastructure)	FR-1.1 through FR-1.4 describe recording transactions on the blockchain, but there is no requirement addressing failure scenarios (e.g., blockchain network unavailable, transaction timeout, consensus failure).
QC-ERR-006	Authorship	Section 1.4.3 System Boundaries	The section number 1.4.3 appears twice (pages 9 and 10) with different content, indicating a numbering or structural error.

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Error ID	Error Type	Location	Characteristic of Error
QC-ERR-007	Omission	Section 3.4 (Product Verification)	FR-4.1 allows customers to retrieve product lifecycle trace, but no requirement specifies system behavior when a scanned QR code is invalid, tampered, or not found in the blockchain.
QC-ERR-008	Contradiction	Section 1.4.1 (Performance Requirements) vs Section 5.1 NFR-2	Section 1.4.1 specifies blockchain transaction confirmation time as "<30 seconds for standard transactions". NFR-2 requires confirmation "within 10 seconds for standard operations". These targets are inconsistent.
QC-ERR-009	Authorship	Section 5.5.3 QF-3 Tamper Resistance, Measurable Criteria	The criterion "100% enforcement of role-based access control" is not measurable as written; it lacks a testable definition (e.g., how enforcement is verified).
QC-ERR-010	Omission	Section 6.5 (Migration & Transition)	While migration tools are mentioned, there are no functional requirements detailing the data migration process, validation of migrated data, or handling of migration failures.
QC-ERR-011	Contradiction	Section 5.4 NFR-19 vs Section 6.2.5 CM-5	NFR-19 specifies backup RPO $\leq$ 24 hours and RTO $\leq$ 4 hours. CM-5 states RTO $\leq$ 2 hours and RPO $\leq$ 30 minutes. These recovery objectives conflict.
QC-ERR-012	Omission	Entire document	The SRS lacks a glossary or definitions section, leaving many acronyms (HSA, MOH, PDPC, etc.) undefined within the document itself.

Table 5: Detected QC error instances

7 QC Findings

7.1 Finding Summary

Finding ID	Type	Severity	Short Description
QC-FND-001	Contradiction	High	Conflicting statements about MFA requirements (all users vs. only administrators).
QC-FND-002	Contradiction	High	Inconsistent blockchain confirmation time targets (30 seconds vs. 10 seconds).
QC-FND-003	Contradiction	High	Conflicting RPO/RTO values between NFR-19 and CM-5.
QC-FND-004	Omission	High	Missing exception handling for blockchain failures, invalid QR codes, password recovery, and migration failure.
QC-FND-005	Omission	Medium	No glossary or definitions section; acronyms undefined within SRS.
QC-FND-006	Authorship	Medium	Vague or ambiguous terms (e.g., "periodic") and non-measurable criteria.
QC-FND-007	Authorship	Low	Typographical errors (duplicate "shall") and structural inconsistencies (duplicate section heading).

Table 6: Summary of QC findings

7.2 Detailed Findings

QC-FND-001: Contradiction in Multi-Factor Authentication Scope

Related error detection: QC-ERR-001

Type: Contradiction

Severity: High

Context. Section 1.4.1 (In-Scope Deliverables) states that MFA is required for "all user accounts". Section 3.2 FR-2.6 limits MFA to "all administrative accounts".

Result. A developer or implementer cannot determine whether non-administrative users (e.g., suppliers, transporters, customers) must use MFA.

Analysis. These two statements are mutually exclusive; one must be incorrect or the scope must be clarified. This is a latent contradiction that will lead to inconsistent implementation unless resolved.

QC-FND-002: Contradiction in Blockchain Transaction Confirmation Time

Related error detection: QC-ERR-008

Type: Contradiction

Severity: High

Context. Section 1.4.1 (Performance Requirements) specifies blockchain transaction confirmation as "<30 seconds for standard transactions". Section 5.1 NFR-2 requires confirmation "within 10 seconds for standard operations".

Result. Two different performance targets exist for the same operation, creating ambiguity for design and testing.

Analysis. The SRS provides conflicting non-functional requirements. The customer must decide which target is correct.

QC-FND-003: Contradiction in Recovery Objectives

Related error detection: QC-ERR-011

Type: Contradiction

Severity: High

Context. NFR-19 (Section 5.4) sets $RPO \leq 24$ hours and $RTO \leq 4$ hours. CM-5 (Section 6.2.5) sets $RTO \leq 2$ hours and $RPO \leq 30$ minutes.

Result. These values are significantly different and cannot both be true.

Analysis. The SRS contains contradictory disaster recovery targets, which will confuse architects and operators.

QC-FND-004: Omission of Critical Exception and Edge Cases

Related error detection: QC-ERR-003, QC-ERR-005, QC-ERR-007, QC-ERR-010

Type: Omission

Severity: High

Context. The SRS specifies normal operational behaviour but omits several important exception scenarios:

- No password reset/recovery process despite account lockout.
- No handling of blockchain unavailability or transaction failure.
- No behaviour defined for invalid or tampered QR codes during product verification.
- No detailed requirements for data migration validation or failure handling.

Result. Downstream design and implementation may fill these gaps with assumptions, leading to inconsistent or incorrect system behaviour in edge cases.

Analysis. These omissions represent significant completeness risks, particularly in a regulatory and safety-critical domain.

QC-FND-005: Missing Glossary

Related error detection: QC-ERR-012

Type: Omission

Severity: Medium

Context. The SRS contains numerous acronyms (HSA, MOH, PDPC, IMDA, GMP, RPO, RTO, etc.) but provides no glossary or definitions section.

Result. Readers unfamiliar with the domain may misinterpret requirements.

Analysis. Including a glossary is a standard practice to ensure clarity and shared understanding.

QC-FND-006: Ambiguous and Non-Measurable Requirements

Related error detection: QC-ERR-004, QC-ERR-009

Type: Authorship

Severity: Medium

Context. Several requirements use vague or undefined terms:

- FR-3.9: "periodic GPS updates" – frequency not specified.
- QF-3 Measurable Criteria: "100% enforcement of role-based access control" – lacks testable definition.

Result. These requirements cannot be verified objectively and may be interpreted differently by different stakeholders.

Analysis. The intended meaning is clear to the authors, but the written statements lack precision needed for implementation and testing.

QC-FND-007: Editorial and Structural Inconsistencies

Related error detection: QC-ERR-002, QC-ERR-006

Type: Authorship

Severity: Low

Context. Minor errors reduce document professionalism:

- Duplicate "shall" in NFR-4 and NFR-7.
- Duplicate section heading 1.4.3 appearing twice with different content.

Result. While these do not affect requirement meaning, they detract from readability and suggest insufficient editorial review.

Analysis. Such errors are typical of early-stage documents and can be easily corrected.

8 Conclusion

The QC audit concludes that the Software Requirements Specification is substantively mature and reflects strong work carried out during the earlier Requirements Engineering stages. However, the document still contains defects typical of the validating stage, particularly in the areas of contradictions (MFA scope, performance targets, recovery objectives), omissions (exception handling, recovery processes, glossary), and authorship clarity (vague terms, editorial errors).

The QC team does not directly modify the SRS. Instead, the findings documented in this report are presented for customer consideration. The customer will determine what actions, if any, should be taken to address the identified defects.

Overall, the SRS is assessed as **conditionally acceptable**, pending customer review and resolution of the findings.

9 Sign-off

9.1 QC Audit Team Authentication

Name	Role	Authentication
Muhammad Ridwan Putra Jasni	QC Audit Lead	Submitted by named team member
Lin Yuhao	QC Auditor	Submitted by named team member
Muhammad Mikhail Bin Mazlan	QC Auditor	Submitted by named team member
Kannan s/o Rajamohan	QC Auditor	Submitted by named team member

Table 7: QC audit team sign-off

10 Addenda

10.1 Repository Information

PROJECT REPOSITORY https://github.com/Spooky312/The-Abyss-Navigation-Suite-INF2113 <i>QC report, SRS, models, and traceability artefacts maintained under version control</i>
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10.2 Glossary

Term	Definition
QC	Quality Control inspection of a work product, in this case the SRS.
SRS	Software Requirements Specification used as the requirements baseline for downstream development.
Authorship defect	Defect caused by unclear or imprecise expression of intended requirement meaning.
Omission defect	Defect in which required content, scenario, or supporting information is missing.
Contradiction defect	Defect in which different parts of the SRS imply inconsistent or conflicting behaviour.
Traceability coherence	The degree to which requirements, models, sources, and references link together clearly and consistently.
MFA	Multi-Factor Authentication.
RPO	Recovery Point Objective – maximum acceptable age of files restored from backup.
RTO	Recovery Time Objective – maximum acceptable downtime.

Table 8: QC glossary addendum

11 AI Review

The following prompts were used with an AI assistant during the QC audit process to assist in identifying potential defects. The responses provided by the AI were considered alongside manual inspection.

Prompt 1

"Review the attached SRS for contradictions in security requirements, particularly around multi-factor authentication (MFA)."

AI Response:

In Section 1.4.1 (In-Scope Deliverables), under "Authentication & Security", it states: "Multi-factor authentication (MFA) for all user accounts with SMS/email OTP support." However, Section 3.2 FR-2.6 says: "The system shall enforce multi-factor authentication (MFA) for all administrative accounts." This is a direct contradiction: the first requires MFA for all users, the second only for administrators. This inconsistency needs resolution.

Prompt 2

"Identify any missing exception-handling requirements in the blockchain transaction recording functional requirements."

AI Response:

FR-1.1 to FR-1.4 describe recording transactions, validating hash integrity, executing smart contracts, and viewing records. There is no requirement addressing failures such as blockchain network unavailability, transaction timeout, consensus failure, or retry logic. These omissions could lead to undefined system behavior in degraded modes.

Prompt 3

"Point out any vague or undefined terms in the functional requirements."

AI Response:

FR-3.9 states: "The system shall allow Transporters to submit periodic GPS updates during shipment transit." The term "periodic" is not defined; it could mean every minute, every hour, or upon specific events. This ambiguity should be resolved by specifying a frequency or trigger condition.

Prompt 4

"Check for inconsistencies in performance targets across the document."

AI Response:

Section 1.4.1 (Performance Requirements) states blockchain transaction confirmation should be "<30 seconds for standard transactions". Section 5.1 NFR-2 requires confirmation "within 10 seconds for standard operations". These are conflicting values and must be aligned.

Prompt 5

"Identify any typographical or formatting errors."

AI Response:

In Section 5.1, NFR-4 and NFR-7 both contain a duplicated word: "shall shall complete". Also, Section 1.4.3 appears twice (pages 9 and 10) with different content, indicating a numbering error.

References

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